

Fundamentals of Strength of Materials – 2018-2019

Tutorial #6: Bending

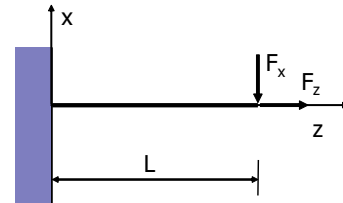
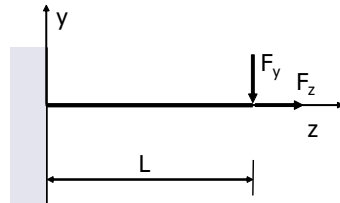
Example #1

A clamped beam is loaded as pictures show. For each cross section herein depicted: draw the neutral axis, indicate the point where maximum stress applies, compute the stress components and the safety factor F_s at that point, for the following loading conditions:

1) $F_y = 5000 \text{ N}$; $F_x = 0 \text{ N}$; $F_z = 60 \cdot 10^3 \text{ N}$;

2) $F_y = 5000 \text{ N}$; $F_x = 2000 \text{ N}$; $F_z = 60 \cdot 10^3 \text{ N}$.

$L = 2 \text{ m}$



$b = 60 \text{ mm}$;

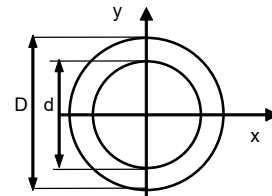
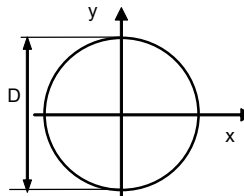
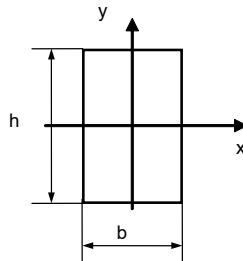
$h = 100 \text{ mm}$;

$D = 100 \text{ mm}$;

$d = 70 \text{ mm}$.

Material:

Steel, $S_y = 315 \text{ MPa}$



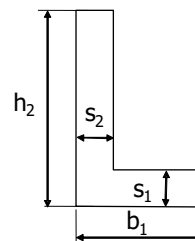
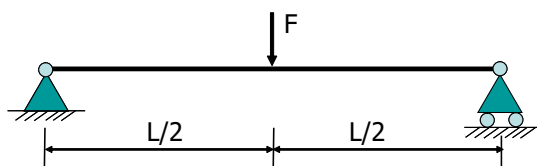
[1) Rectangle: $F_s = 2.8$. Solid Circle: $F_s = 2.8$. Hollow Circle: $F_s = 2.1$.

2) Rectangle: $F_s = 1.8$. Solid Circle: $F_s = 2.7$. Hollow Circle: $F_s = 1.9$]

Example #2

For the steel structure shown in figure, compute the bending moment acting on the most loaded cross section, assumed to be the same of Tutorial 5 – Example #3.

$F = 10 \cdot 10^3 \text{ N}$ $L = 1 \text{ m}$



$b_1 = 80 \text{ mm}$

$h_2 = 200 \text{ mm}$

$s_1 = 13 \text{ mm}$

$s_2 = 10 \text{ mm}$

Find the centroid location, draw stress applied to the cross section and show the neutral axis.

Identify the most stressed point(s) on the cross section and compute the related value of normal stress σ_{zz} into the principal reference frame, (x, y) .

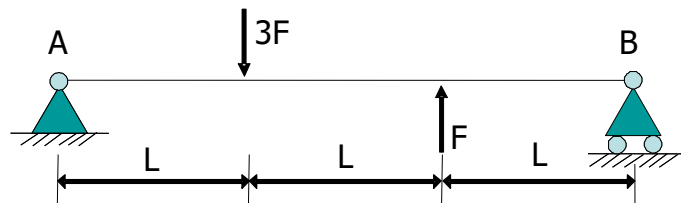
Compute the safety factor F_s at the most stressed point. Material: steel, yield strength $S_y = 600 \text{ MPa}$.

Draw the Mohr's circle for a) the point of maximum tensile stress and b) the point of maximum compressive stress.

[a) $\sigma_{zz} = 31.5 \text{ MPa}$; b) $\sigma_{zz} = -35 \text{ MPa}$; $F_s = 6.7$]

Example #3

A shaft with circular cross section of diameter d , whose material is steel 50CrMo4, is loaded as shown in figure. Compute the value of diameter d being compatible with a safety factor $F_s = 1.5$. Refer to the table below for inputs about the mechanical properties of material.



$F = 10 \text{ kN}$
 $L = 150 \text{ mm}$

[$d = 37 \text{ mm}$]

	$d \leq 16 \text{ mm}$				$16 < d \leq 40 \text{ mm}$				$40 < d \leq 100 \text{ mm}$				$100 < d \leq 160 \text{ mm}$			
	S_y	S_u	A	q	S_y	S_u	A	q	S_y	S_u	A	q	S_y	S_u	A	Q
	N/mm ²		%		N/mm ²		%		N/mm ²		%		N/mm ²		%	
50CrMo4	900	1100	9	40	780	1000	10	45	700	900	12	50	650	850	13	50

S_y : yield strength;

S_u : ultimate strength;

A : strain at fracture;

q: the reduction of area at fracture (initial length between reference marks $L_0 = 5,64 \sqrt{A_0}$)

Example #4

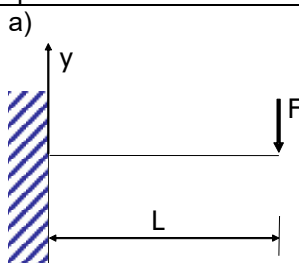
Compute the expression of elastic line of beams (material: steel, circular cross section with $D=100 \text{ mm}$) shown in the following pictures. Compute also the transverse displacement at the free end of the beams.

Data:

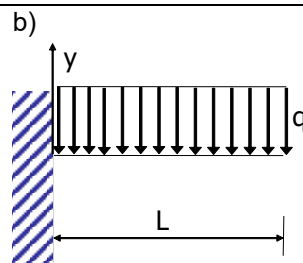
$F = 5000 \text{ N}$;

$L = 2 \text{ m}$;

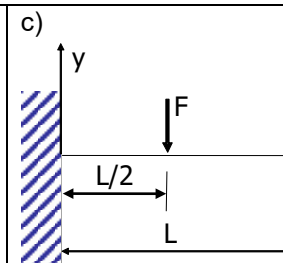
$q = 2.5 \text{ N/mm}$



$\alpha(z): C = 0; v(z): D = 0$



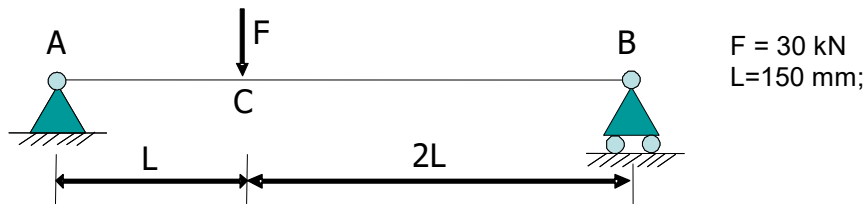
$\alpha(z): C = 0; v(z): D = 0$



$\alpha_I(z): C_I = 0; v_I(z): D_I = 0$
 $\alpha_{II}(z): C_{II} = FL^2/8EI_{xx}; v_{II}(z): D_{II} = FL^3/48EI_{xx}$

Example #5

For the shaft herein depicted (circular cross section, $d = 30$ mm, steel 50CrMo4) compute the expression of deflected shape and find the value of rotation angle α at points A and B, where rolling bearings are located, and vertical displacement of point C.



$$C_I = 5FL^2/9EI_{xx}; D_I = 0; C_{II} = 19FL^2/18EI_{xx}; D_{II} = FL^3/6EI_{xx}.$$